Simulating and Testing the Biomarker-by-Drug Interaction in the Hybrid Design: Where We Are

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Author. pmsimstats team. **Scope.** The recommended data-generating process (DGP), called the refined process, the recommended analysis, and the evidence for each. Details: `docs/36` to `docs/38`, `docs/44` to `docs/49`, and paper 01. **Symbols.** c_{bm} biomarker coupling to the drug response; $t_{1/2}$ carryover half-life (weeks); t_{sd} time since discontinuation; ρ AR(1) correlation per week; c_1 cross-component correlation; N total sample size.

1. The data-generating process

Starting point. Hendrickson et al. (2020; "RH") simulate the score as baseline minus three Gompertz responses: drug (BR), expectancy (PB) and natural course (TV). The biomarker is bound to the drug response as a *correlation* c_{bm} in a joint multivariate normal, a parameterization new to the N-of-1 literature to our knowledge. Two defects follow from the published implementation.

Defect in RH's code	Consequence	Evidence	Fix
Biomarker coupled at full strength wherever any drug effect remains (step rule)	interaction vanishes at any positive half-life	Hybrid, $N = 70$: power 0.66 to 0.08 at $t_{1/2} = 0.1$ (RH's code, 250 replicates)	**graded coupling**: c_{bm} on drug, $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug, 0 before exposure; power 0.62 to 0.68
Compound symmetry allows c_{bm} only below 0.256; larger values silently repaired	RH's $c_{bm} = 0.6$ results describe a different process	20 of 54 matrices repaired	**separable AR(1)** ($\rho = 0.7$): ceiling 0.48; **stop, do not repair**
Carryover recursion compounds the decay	off-drug means too small	code inspection	**anchored carryover** from the last on-drug value

How we got there. First, closed-form power for a paired-difference statistic showed that power depends on carryover only through the *coupling gap* (on minus off coupling). The step rule sets that gap to 0; graded coupling lets it fall smoothly, and provably monotonically (docs/37). Next, the eight-line fix restored power in RH's own code. Then the separable form was shown to be valid for every visit schedule, to give an exact ceiling as a sum of per-switch costs, and to match the random-intercept-plus-AR(1) analysis model (docs/35).

Key insight. Conditional on the biomarker, covariance binding is mean moderation with a smaller residual variance. How much carryover costs therefore depends on whether the interaction fades with the drug effect, not on where it is bound. Mean binding with the same fading profile loses power almost identically: 0.454 against 0.473 at one week.

The refined process (recommended DGP).

- **Core:** graded coupling, separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, anchored carryover, a stop on infeasible matrices.
- **Re-exposure (retained from the published process):** the drug-response curve restarts at zero, as in Hendrickson's code; this is consistent with re-titration of prazosin.
- **Carryover range:** pharmacological half-life 0.1 to 0.2 week.
- **Options:** prognostic biomarker couplings (natural course, placebo, baseline severity), phase-specific measurement error, Weibull decay, dropout.

Mean binding, or the drug-sensitivity model of `docs/38` (not yet implemented), covers effects above the ceiling.

2. The analysis

How we got there. Three rounds of simulation, each on identical trials, with size judged before power:

1. **Covariance study** (`docs/45`, 378,000 fits). Kenward-Roger corrects an unstructured covariance; CR2 corrects a parsimonious one. Neither substitutes for the other.
2. **Exploratory stress test** (`docs/46`, 41 cells, 28,000 trials, 11 protocol analyses). It varied correlation structure, N, carryover, decay form, dropout and measurement error, and added a *prognostic* biomarker.

3. **Post hoc arms** on the same trials, judged by the same rule (`docs/46`, Amendment 2): phase-specific variances with the expectancy terms (G1, G2), AIC selection with them (F1A), and two baseline-severity cells.

The decisive finding concerns analyses that keep baseline as an outcome row. They share one biomarker slope between baseline and the off-drug visits. If the biomarker also predicts the natural course or the placebo response, part of that association is reported as a drug interaction (`docs/48`, `docs/49`).

Analysis (random intercept + CAR(1) residuals unless stated)	Size, standard cells	Size, prognostic cells	Power, AR(1) $c_{bm} = 0.3$	Power, CS $c_{bm} = 0.25$
Published (`mmrm` compound symmetry, Satterthwaite df)	0.069 (0.081 at $N = 35$)	0.053 to 0.159	0.398	0.596
Unstructured, Kenward-Roger	0.057 (0.087 at $N = 35$)	0.195 to 0.416	0.534	0.618
S: fixed half-life 0.5, CR2	0.048	0.049 to 0.100	0.514	0.554
DS: S with AIC-selected half-life	0.053	0.058 to 0.115	0.486	0.480
C1: S with phase-specific variances	0.053	0.173 to 0.382	0.578	0.660
F1: S plus expectancy terms `De + bmc:De`	**0.051**	**0.040 to 0.047**	**0.364**	**0.522**
G1: F1, phase variances, CR2 (post hoc)	0.049	0.042 to 0.055	0.354	0.546
G2: G1, model-based (post hoc)	0.047	0.043 to 0.062	0.380	0.528
F1A: F1, AIC half-life (post hoc)	0.057 (0.073 at $t_{1/2} = 1$)	0.047 to 0.059	0.338	0.446

Size: null rejection, nominal 0.05 (standard cells pooled over 11,000 trials; band 0.046 to 0.054). Power is size-adjusted, 500 trials per cell. "Prognostic": the biomarker correlated 0.3 with the natural course (constant or growing) or with the placebo response. In a baseline-severity null cell every analysis is within the 0.075 cell limit (S 0.050, F1 0.040).

Reading the table.

- **Phase-specific variances (C1)** are the most powerful, but only because they weight the shared baseline slope heavily. Once the baseline is freed (G1, G2), the power advantage disappears and the robustness is kept. On maximin shortfall G2 (0.232) and F1 (0.242) tie within 0.02, and the rule gives the tie to F1, which has fewer covariance parameters.
- **AIC selection (DS)** holds its size and estimates the carryover half-life sensibly: it picks 0 82% of the time when there is none, and 1 or 2 weeks 86% of the time when it is 1. It costs about 0.03 power against a fixed 0.5 week. It does not protect against a prognostic biomarker by itself, and with the expectancy terms (F1A) it fails the size criterion.
- **The expectancy terms are what make an analysis robust.** Their cost is about 30% of power under strong serial correlation, and small under compound symmetry.

Recommended analysis.

- **Primary:** F1. Random intercept with `corCAR1` residuals, CR2 standard errors, linear time, baseline as an outcome row, the exposure-decayed drug indicator at a fixed half-life of 0.5 week (not AIC-selected, since F1A fails size), and `De + bmc:De`.
- **Sensitivity analyses:** S (biomarker assumed purely predictive), the freed-baseline version, and the binary drug indicator. Disagreement between F1 and S is evidence of a prognostic biomarker.
- **Sample size:** plan it for F1, stating the serial-correlation assumption.

Open items. The refinements still have to move into the package, and its sampler's silent repair has to go. The covariance study and the stress test ran configuration C of `docs/36` at $\rho = 0.7$, the covariance construct of the refined process, with the published recursive carryover and fixed allocation, so the refined process with anchored carryover has not itself been simulated (inspected in `08-covariance-study.R` and `11-strawman-stress-test.R`). Effect sizes are not yet calibrated to prazosin data: blood pressure's placebo-arm association was null in only 35 participants (Raskind et al. 2016). Any post hoc choice (G1, G2) needs an independent run.